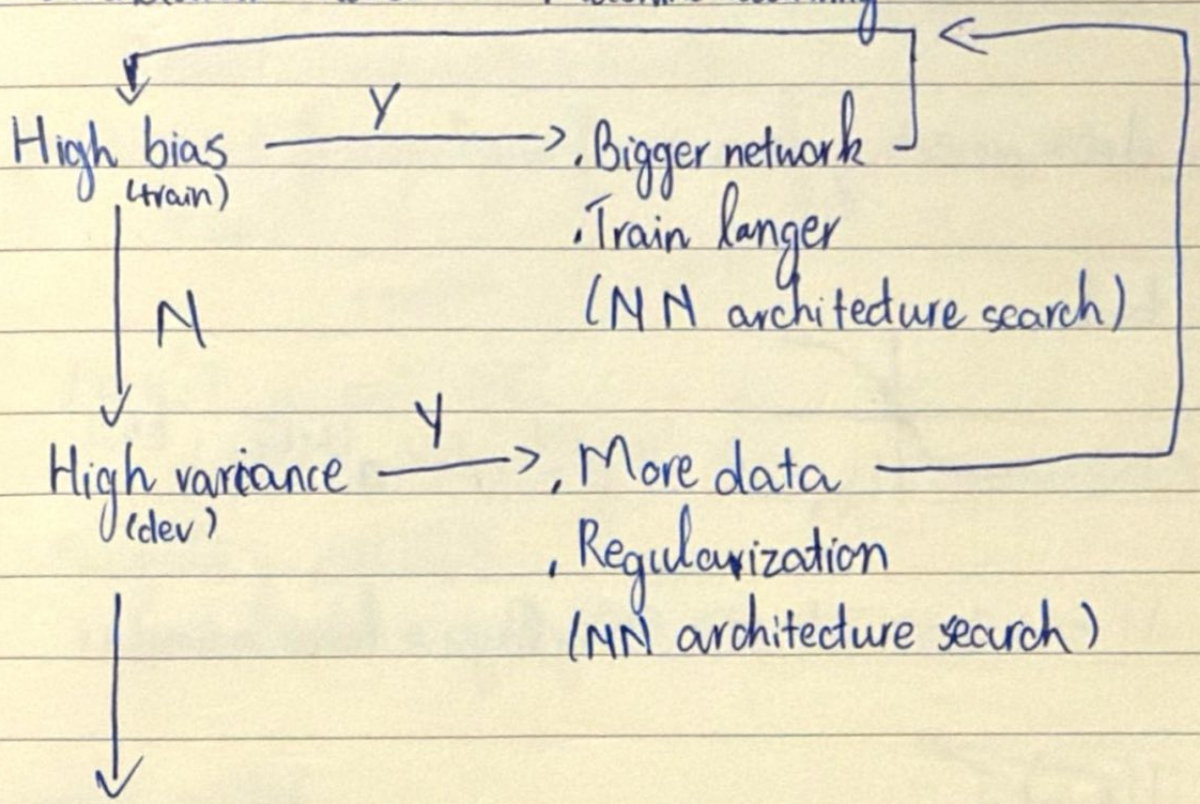
DATE / / TITLE

Preprocess for DL

1. Figure out the dimensions and shapes (m_train, m_test, num_px, ...)
2. Reshape the dataset (numpx * numpx * 3, 1)
3. Standardize data

Machine learning



Done!!

Machine learing regularization

L2 regularization: $||w||_2^2$, Euclid (popular)

L1 regu: $\frac{\lambda}{2m} ||w||_1$; w will be be sparse

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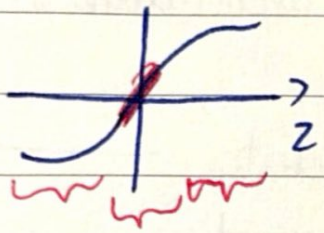
NN regu.:

- weight decay = Frobenius norm = L2 regu

↳ $\left(1-\frac{\alpha\lambda}{m}\right).w^{[l]}$

How does preventing from over fitting?

Ex: tanh



$z^{[l]} = w^{[l]}.a^{[l-1]} + b^{[l]}$

$\lambda\uparrow \rightarrow w\downarrow \rightarrow z\downarrow \Rightarrow$ every layer ≈ linear → simpler

~~J(b,...) =~~

In iteration plot: if there is regul: ~~cost~~ lost function ↓

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- dropout regularization :

Why no zeros initialize ?

- since the weights and bias = 0, create vector that is 0

$$Relu(z) = max(0, z) = 0$$

- for sigmoid function :

$$\sigma(z) = \frac{1}{1+e^{-(z)}} = \frac{1}{2} = y\text{-pred}$$

For $y = 1$, $y_pred = 0,5$:

$$L(0,1) = -(1)\ln\left(\frac{1}{2}\right) = 0,693147180559945\overline{3}$$

For $y = 0$, $y_pred = 0,5$:

$$L(0,0) = L(0,1) = 0,301029$$

Improving model :

- Random Initialize (small, no 0, symetric breaking)
- L2
- Propout

$$loss_BCE = -(y \log \hat{y} + (1-y)\log(1-\hat{y}))$$

$y = 0,1$

$\hat{y} = 0,05$

$0,3457$

$0,1 \sim -2,3$

$0,0005 \sim -7,6$

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Gradient checking:

Caculate ~~grad~~ gradapprox:

1. $\theta^+ = \theta + \varepsilon$
2. $\theta^- = \theta - \varepsilon$
3. $J^+ = J(\theta^+)$
4. $J^- = J(\theta^-)$
5. $\text{gradapprox} = \frac{J^+ - J^-}{2\varepsilon} \approx 10^{-7}$

Adam optimization:

Combine: RMSpop + momentum

Hyperparameter choice:

α: needs to be tune

β_1: 0,9

β_2: 0,999

ε: 10^{-8}

Learning rate decay:

$$\alpha = \frac{1}{1 + \text{decay-rate} * \text{epoch-num}} \alpha_0$$

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Mini-batches:

Shuffling
Partitioning
} build mini-batches

- size mini-batch = power of 2 (16, 32, 64, ...)

Momentum: Average of the deriative of last 10 epochs to update
- Tune β, learning-rate α

RMSprop: Reduce the update in the wrong direction.

~~Adam~~:

Hyperparameters

α

β

$\beta_1, \beta_2, \varepsilon$

layers

hidden units

learning rate decay

mini-batch size

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Batch norm:



Loss: Crossentropy (no normalized)
↳ Kocho qua softmax
NLLLoss: choqua Log softmax

Chain of assumption in ML:

Fit training set ⟷ bigger network, Adam
↓
Fit dev set ⟷ Regularization, Bigger training set
↓
Fit test set ⟷ Bigger dev set
↓
Perform well in real world ⟷ change dev set, cost func

Metric:

- Human performance
- Accuracy, F1

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	General speech Regcognition	Rearview mirror speech data
Human level	"Human level" 4%	
	| avoidable bias	
Error an examples trained on	"Training error" 7%	
	| variance	
Error on examples not trained	"Training-dev' 10% error"	"Dev/Test 6% error"
	— data mismatch	

Addressing data mismatch

- Carry out manual error analysis to try to understand difference between train/dev/test set
- Making training data more similar; or collect more data similar to dev/test sets

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- Universal Approximation Theorem :

"NN with a certain structure can approximate any continuos function to any desired degree of accuracy"

"Black box" cause nobody knows how to conduct a (good) model, no instruction, no nothing

Transfer learning (fine tune)

. Task A and B are the same input
. You have a lot more data for task A than task B
. low level features from A could be helpful for learning B

If task B's data is low -> finetune the last layer
large -> retrain the whole model

Multi-task learning

. Traing an a set of tasks that could benefit from having shared lower-level features.
. Usually: amount of data you have for each task is quite similiar
. Can train a big enough NN to do well on all tasks.

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End-to-end deep learning

Pros:

. let the data speak

. less hand-designing of components needed

Cons:

. May need lots of data

. Excludes potentially useful hand-designed components

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Tại sao dùng sigmoid mà ko dùng RELU?

Tại sao tăng dần mà không đều, giảm dần giảm feature map

bỏ bỏ

nhiều -> overfit thừa

Đặc trưng nào là hiệu quả nhất -> đen, trắng

làm sao pooling mà vẫn đảm bảo chất lượng của ảnh?
=> càng giảm càng giảm kích thước nhưng vẫn giữ được đặc trưng quan trọng, ↓ tham số ↓ overfitting

Tại sao CNN lại dùng pool ~~boxy~~ mà không dùng for

Tại sao dùng max mà ko dùng avg pool?
- Dễ bị noise -> GD nhanh hơn
- Giá trị thật

Avg -> visualize

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